

Vacuum Cleaner Problem

1. Aim

To implement an **intelligent vacuum cleaner agent** that can perceive the condition of rooms and perform suitable actions to clean all the dirty locations efficiently.

2. Problem Statement

The Vacuum Cleaner problem is a simple Artificial Intelligence problem in which an intelligent agent moves between two rooms, **Room A and Room B**, and cleans them whenever they are dirty. The agent must observe the current location and room condition and decide whether to clean the room or move to the other room.

3. Solution

The problem is solved using a **rule-based intelligent agent**. The agent checks its current location and the cleanliness of the room. If the current room is dirty, it performs the **Suck** action to clean it. If the room is already clean, the agent moves to the other room. The process continues until both rooms are clean.

4. Algorithm

1. Start the vacuum cleaner in either Room A or Room B.
2. Check the current room's condition.

3. If the current room is dirty, perform the **Suck** action.
4. If the current room is clean, move to the other room.
5. Check the condition of the new room.
6. Clean the room if it is dirty.
7. Continue the process until both rooms are clean.
8. Display the actions performed by the vacuum cleaner.
9. Stop when all rooms are clean.

5. Program

```
def vacuum_cleaner(room_a, room_b, position):  
    rooms = {  
        "A": room_a,  
        "B": room_b  
    }  
  
    print("Initial State:")  
    print("Room A:", rooms["A"])  
    print("Room B:", rooms["B"])  
    print("Vacuum Position:", position)  
    print()
```

```
while "Dirty" in rooms.values():

    if rooms[position] == "Dirty":
        print("Vacuum is in Room", position)
        print("Action: Suck")
        rooms[position] = "Clean"

    else:
        if position == "A":
            position = "B"
            print("Action: Move to Room B")
        else:
            position = "A"
            print("Action: Move to Room A")

print("\nFinal State:")
print("Room A:", rooms["A"])
print("Room B:", rooms["B"])
print("Both rooms are clean.")
```

```
# Initial conditions
```

```
room_a = "Dirty"
```

```
room_b = "Dirty"
```

```
position = "A"
```

```
vacuum_cleaner(room_a, room_b, position)
```

6. Result

The Vacuum Cleaner intelligent agent was successfully implemented using Python. The agent correctly identifies dirty rooms, performs the **Suck** action to clean them, and moves between rooms when required.

Output:

Initial State:

Room A: Dirty

Room B: Dirty

Vacuum Position: A

Vacuum is in Room A

Action: Suck

Action: Move to Room B

Vacuum is in Room B

Action: Suck

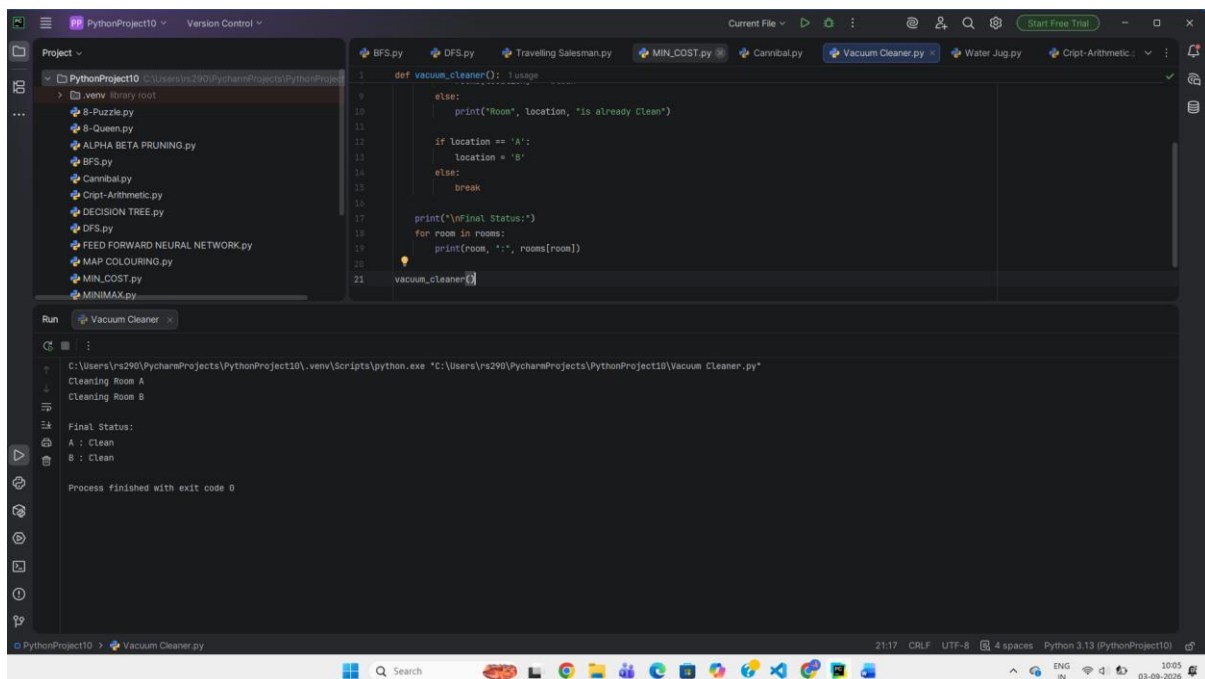
Final State:

Room A: Clean

Room B: Clean

Both rooms are clean.

Thus, the intelligent vacuum cleaner successfully cleaned **both Room A and Room B** by selecting appropriate actions based on the current state of the environment.



The screenshot displays a Python IDE interface for a project named 'PythonProject10'. The left sidebar shows a file explorer with various Python scripts, including 'Vacuum Cleaner.py'. The main editor window shows the code for the 'vacuum_cleaner' function, which iterates through rooms and prints their status. The bottom panel shows the output of the program, indicating that both Room A and Room B are clean.

```
def vacuum_cleaner():  
    1 usage  
    2  
    3 else:  
    4     print("Room", location, "is already clean")  
    5  
    6     if location == 'A':  
    7         location = 'B'  
    8     else:  
    9         break  
    10  
    11     print("\nFinal Status:")  
    12     for room in rooms:  
    13         print(room, ":", rooms[room])  
    14  
    15     vacuum_cleaner()  
    16
```

Run Vacuum Cleaner

```
C:\Users\rs290\PycharmProjects\PythonProject10\.venv\scripts\python.exe "C:\Users\rs290\PycharmProjects\PythonProject10\Vacuum Cleaner.py"  
Cleaning Room A  
Cleaning Room B  
Final Status:  
A : Clean  
B : Clean  
Process finished with exit code 0
```